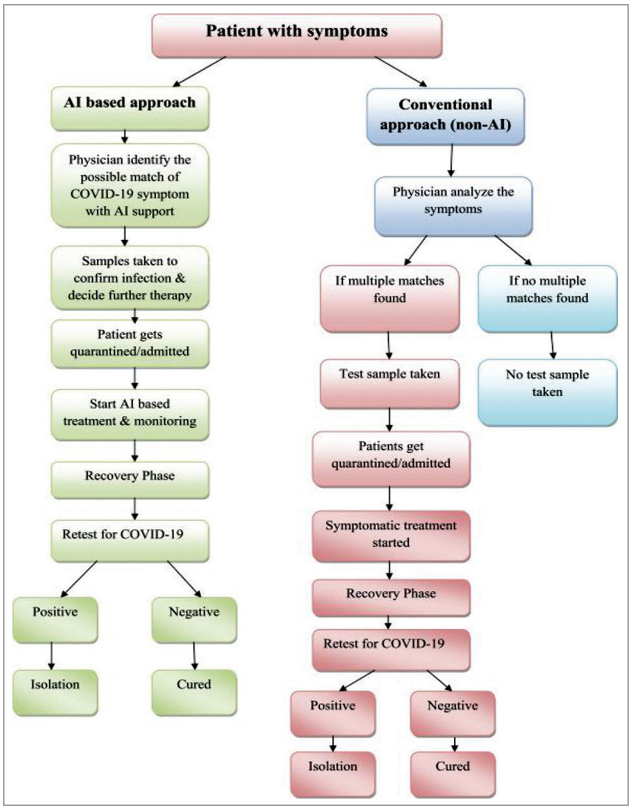
Application of GIS to Covid-19

Geographic understanding is essential in detecting, understanding, and responding to pandemics such as Covid-19. A geographic information system (GIS) is a computer system for capturing, storing, checking, and displaying data related to positions on the earth's surface in a meaningful way through interactive maps or other infographics. GIS techniques and tools are useful in finding and monitoring large-scale population-based occurrences such as disease clusters, outbreaks of infection, and possible associations between infections and environmental factors.



Flow chart comparing general procedure of AI and non-AI followed to identify Covid-19 symptoms along with treatments needed (Credit: www.sciencedirect.com)

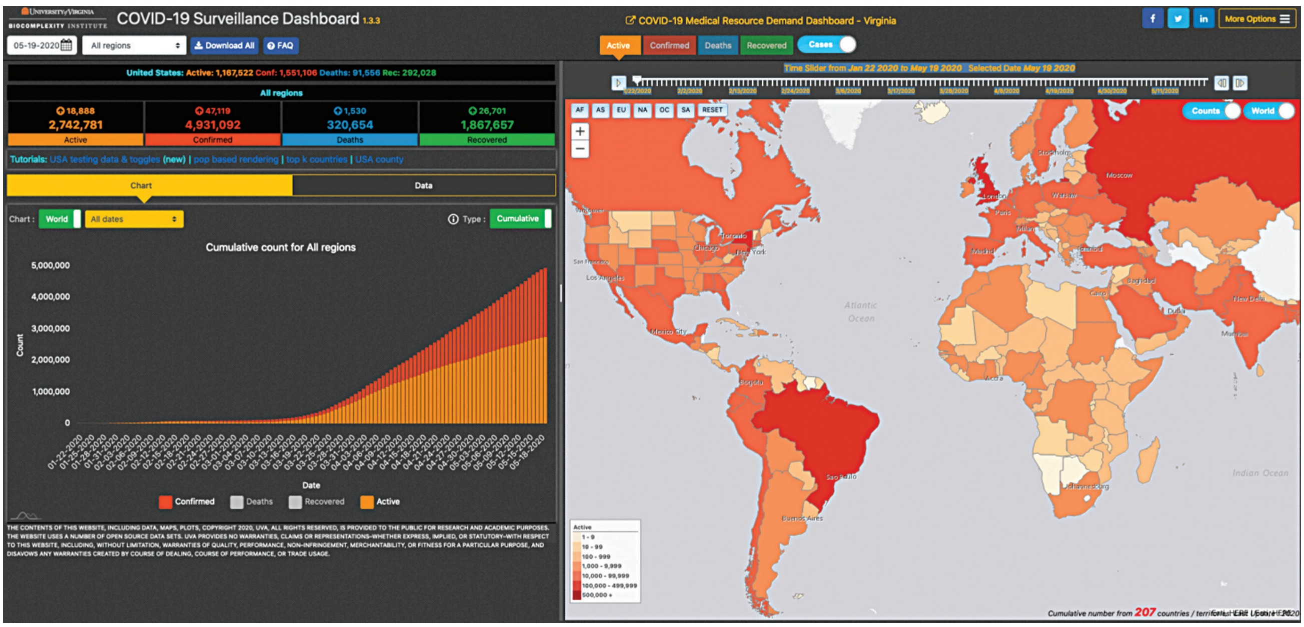
GIS helps public health agencies, policymakers, and administrators to map disease occurrence against multiple parameters such as demographics, environment, geographies, and past occurrences to understand the origin of an outbreak, its spread pattern, and intensity. GIS enables identifying at-risk populations, planning targeted intervention, evaluating facilities and healthcare capacities, and establishing effective communication among supporting agencies to ensure a coordinated response.

It helps epidemiologists study the outbreak of diseases and how the incidence of congenital diseases varies over space and time. WHO, Red Cross, academics, local governments, and hospitals employ GIS to communicate spatial data sets to the public and help policymakers to prepare and allocate resources appropriately.

The GIS dashboard app, created by John Hopkins University, Baltimore, allows users to click on a country to identify confirmed cases and deaths. Another app, Covid-19 Surveillance Dashboard, created by the University of Virginia, provides a visualisation of Covid-19 cases, recoveries, and deaths across the globe on a real-time basis. Prediction and forecasting systems allow users to identify areas where shortages in medical equipment are likely to reach crisis point and predict surges in the cases to enable better planning for taking pre-emptive actions so that hospitals do not run out of beds, ventilators, masks, and other supplies.

Some more technologies being used to fight Covid-19

Blockchain. As Covid-19 is highly infectious, there is an urgent need for speeding up the detection of virus



Covid-19 Surveillance Dashboard created by the University of Virginia (Credit: https://covid19.biocomplexity.virginia.edu)